

Forecasting Airline Stock Volatility with HAR-RV, IV, OVX, and TOSI

Group Number: 23

Your Names and UT IDS: Blake Stanley (bks2356), Shivsagar Palla (sp56633), Raghuvendra Chowdhry (rbc993)

Analysis: We did not make any changes to our analysis after the presentation. The main analytical change happened after Phase 2: we moved away from the earlier MIDAS/monthly forecasting plan and rebuilt the project around a HAR-style realized-volatility framework. In the final version, we computed realized volatility from 5-minute Alpaca data, combined HAR-RV features with IV, OVX, and TOSI, and evaluated OLS, Ridge, Random Forest, and XGBoost with walk-forward out-of-sample testing. This gave us a cleaner time-series baseline, more direct comparisons across feature sets, and a stronger connection between forecasting results and the trading strategy.

Design:

- Visualization packages used: Plotly for interactive charts and Streamlit for the dashboard interface.
- Interactive daily realized-volatility line charts with range sliders, shaded periods, and hover tooltips.
- An oil-driver view that overlays airline volatility with OVX and TOSI so users can inspect co-movement over time.
- Monthly correlation heatmaps and scatterplot facets for IV, OVX, and TOSI against future airline volatility.
- Model-comparison visuals for RMSE, Sharpe ratio, and Diebold-Mariano significance across feature sets and model families.
- Forecast-vs-actual panels, cumulative JETS straddle P&L; charts, and feature-importance bars for the tree-based models.

Strengths:

- The project has a clear economic story: oil-market uncertainty and sentiment should matter for airline volatility.
- We used multiple data sources and frequencies, including intraday realized volatility, options-implied volatility, OVX, and TOSI.
- The HAR baseline made the modeling framework interpretable, and the feature-addition design let us test what each signal contributed.
- Our evaluation was disciplined: walk-forward out-of-sample testing, per-ticker comparisons, and trading-metric validation.

- The final Streamlit app made the project easier to explain because users could interact with the evidence instead of only seeing static charts.

Challenges:

- Aligning mixed-frequency data was difficult because realized volatility and IV are daily/intraday while TOSI is monthly.
- Several datasets had missing or uneven coverage, especially older implied-volatility history, which required careful cleaning and alignment.
- The external oil signals were informative, but improvements were not uniform across every airline, so interpretation had to stay nuanced.
- Running walk-forward experiments across several model families and feature sets was computationally expensive and required saving artifacts carefully.

Advice: Start the data pipeline early and lock down your target variable before you build models or visuals. Choose a strong, interpretable baseline first, because it becomes much easier to justify added complexity later. Also separate expensive modeling from the presentation layer by saving clean artifacts for the dashboard. Finally, do not rely only on correlations or in-sample fit; use true out-of-sample testing and keep your narrative tied to a concrete decision problem.